

Andy Duong

AI Backend Engineer / AI Workflow Engineer

Python & FastAPI · Agentic AI & RAG · Async Workflows · API Integrations · Production Delivery

khangss2803@gmail.com · +84 835 608 819 · WhatsApp · LinkedIn · GitHub · Can Tho, Vietnam (UTC+7)

SUMMARY

AI backend engineer focused on production APIs, agentic workflows, RAG, real-time voice AI, document processing, and third-party integrations. Currently working at **FPT Software**, a global technology and IT services company operating across 30+ countries, with hands-on enterprise training in AI engineering and AI-assisted software development. Strong backend foundation in Python/FastAPI and Java/Spring Boot, with practical experience building asynchronous workflows, queues, databases, Dockerized services, CI/CD, and VPS deployments for real business systems.

TECHNICAL SKILLS

AI Backend: Python, FastAPI, Java, Spring Boot, NestJS, REST APIs, WebSocket, SSE, Redis, BullMQ, Celery

AI Workflows: Agentic AI, RAG, LangGraph, LangChain, tool calling, Voice AI, Gemini Live, Azure Speech Services, OCR, Document AI, LLM integration, embeddings, hybrid/vector search, Qdrant, Pinecone

AI Developer Tooling: MCP integrations, reusable agent skills, prompt engineering, context engineering, token-efficient AI workflows, GitHub Copilot

Data & Infrastructure: PostgreSQL, MySQL, Redis, Kafka, Docker/Compose, GitHub Actions, GHCR, Linux VPS, Nginx, Nginx Proxy Manager, Portainer, Cloudflare/R2

Frontend: TypeScript, JavaScript, React, Next.js, HTML, CSS

CERTIFICATIONS & TRAINING

- **GitHub Copilot Certification (GH-300)** — Microsoft / GitHub · Certified · Earned May 2026 · Valid through May 2028
- **Claude Certified Architect, Foundations (CCA-F)** — Anthropic · Certified
- **AI Engineer Program — FPT Software LevelUP** · Completed · Credential
Hands-on curriculum covering OpenAI/Azure OpenAI, function calling, batching/retry, Llama 3, Hugging Face, embeddings, Pinecone, LangChain, RAG, LangGraph, agentic and multi-agent systems, Azure AI, workshops, assignments, and hackathon delivery.

EXPERIENCE

FPT Software

Aug 2025 — Present

Junior Software Engineer · Global Technology & IT Services

- Build and maintain production backend services with **Java, Spring Boot, Python, and FastAPI**, including REST APIs and internal/external service integrations.
- Develop **OCR and AI document-processing workflows** that transform unstructured documents into structured data for downstream business systems.
- Implement **RAG-based knowledge workflows** using chunking, embeddings, vector retrieval, and LLM integration for grounded answers over private data.
- Work across asynchronous processing, databases, Dockerized services, integration testing, deployment, and production support while applying AI-assisted engineering practices to implementation and workflow automation.

TekNix Technology Corporation

May 2025 — Aug 2025

Back End Developer

- Developed REST APIs and backend modules with **NestJS, TypeScript, and PostgreSQL**, covering data models, validation, authentication, authorization, and business logic.
- Integrated backend services with frontend applications and external APIs, with a focus on maintainable service boundaries and reliable data flows.

SELECTED PROJECTS

Agentic Document Chatbot

2026

Independent Project · AI Backend & Workflow Systems

FastAPI · LangGraph · PostgreSQL · Redis/Celery · Qdrant · MinIO/R2 · Docker · GitHub Actions

- Built a secure multi-workspace **Agentic RAG platform** for private document knowledge, supporting PDF, DOCX, TXT, and image ingestion with OCR, chunking, embeddings, indexing, and source-grounded answers.
- Implemented a LangGraph tool-calling workflow with hybrid dense/sparse retrieval, reranking, validated citations, conversation memory, document-aware tools, and streamed responses over SSE.
- Designed asynchronous document processing with Redis/Celery, vector search, S3-compatible storage, workspace isolation, Docker, CI/CD, GHCR, migrations, and VPS deployment workflows.

SujiDo AI Voice Tutor

2026

Product Project · Real-time Voice AI & Learning Automation

Gemini Live Native Audio · Function Calling · Azure Speech · LLM · Redis/BullMQ · TypeScript

- Led the **AI implementation** for a Japanese-learning platform, building a real-time voice tutor that supports Vietnamese/Japanese conversations, microphone audio, live transcripts, VAD, and interruption handling.
- Implemented Gemini Live function calling so the tutor can retrieve verified lesson/progress context and trigger server-validated Japanese pronunciation assessment; used **Azure Speech Services** for speech recognition, text-to-speech, language detection, and pronunciation scoring.
- Designed a controlled AI architecture with structured outputs, schema and semantic validation, allowlisted tools, ephemeral tokens, Redis/BullMQ background jobs, timeout/retry/fallback handling, and privacy-aware telemetry that avoids raw audio and API keys.

The Nail Lounge AI Booking Assistant

2026

Client Project · Agentic AI & Workflow Automation

AI Agent · Tool Calling · External APIs · Redis/BullMQ · Email · WhatsApp · Cloudflare

- Built a customer-facing **tool-calling AI booking assistant** that recommends services, collects customer and appointment details, keeps conversational context, and orchestrates booking workflows through external scheduling APIs.
- Connected the agent to business tools for service/staff/availability lookup, structured lead capture, email actions, booking operations, and automated WhatsApp notifications.
- Used background queues for latency-prone external API and notification work so customer conversations remain responsive while integrations run asynchronously.

Kruzo Document AI

2026 — Present

Independent Product · AI Backend & Document Automation

Python · FastAPI · OCR · LLM · PostgreSQL · Redis · Docker · GitHub Actions

- Built an API-first document automation product that extracts structured data from PDFs and images into **JSON, CSV, and Excel-ready outputs**, including predefined and custom-schema extraction.
- Designed asynchronous extraction workflows that move long-running OCR/LLM and external-provider calls off the request path using Redis-backed background processing.
- Implemented validation, structured error contracts, reusable extraction APIs, containerized deployment, CI/CD, and production hosting for a complete document-processing workflow.

IncodeService.com

Client Product · Backend Systems & Production Delivery

Java · Spring Boot · REST APIs · WebSocket · Docker · GitHub Actions · GHCR · Ubuntu VPS

- Built and operated an automotive-service platform used by **nearly 2,000 users** and handling approximately **100,000 requests per week** across production workflows.
- Designed backend APIs and real-time workflows, exposed integration APIs for a **desktop application**, and managed delivery from implementation to Dockerized CI/CD and VPS deployment.

EDUCATION

Can Tho University

2021 — 2025

Bachelor's Degree in Information Technology · GPA 3.80 / 4.00

English: Working proficiency